

Euclidean Geometry in Mathematical Olympiads

Annotations & Notes

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1 Overview

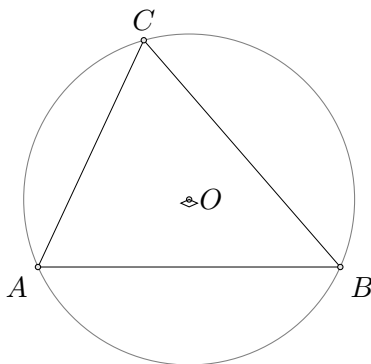
Euclidean Geometry in Mathematical Olympiads (EGMO) by Evan Chen is the standard olympiad geometry reference. It covers the full toolkit: angle chasing, power of a point, radical axes, inversion, projective geometry, and more — each chapter ending with a curated problem set. The writing is unusually clean for a competition math text.

2 Chapter Notes

2.1 Chapter 1 — Angle Chasing

Definition 2.1. The **directed angle** $\angle(AB, CD)$ is the angle of rotation from line AB to line CD , taken modulo 180° .

Theorem 2.2 (Inscribed Angle Theorem). *Let ω be a circle with center O , and let A, B, P lie on ω with P not on arc AB . Then $\angle APB = \angle AOB/2$. Equivalently, all inscribed angles subtending the same arc are equal.*



Remark. Use directed angles to avoid case splits. $\angle APB = \angle AQB$ iff A, P, Q, B are concyclic.

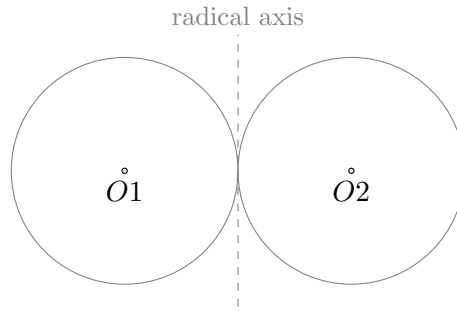
2.2 Chapter 2 — Circles

Definition 2.3. The **power of a point** P with respect to circle ω with center O and radius r is

$$\text{pow}(P, \omega) = PO^2 - r^2.$$

If a line through P meets ω at X and Y , then $\text{pow}(P, \omega) = \overline{PX} \cdot \overline{PY}$ (signed).

Theorem 2.4 (Radical Axis). *The locus of points with equal power with respect to two circles is a line perpendicular to the line through their centers. The radical axes of three circles taken pairwise are concurrent (the **radical center**).*



2.3 Chapter 3 — Lengths and Ratios

Key tools: Stewart's Theorem, mass point geometry, Menelaus, Ceva.

Theorem 2.5 (Ceva's Theorem). *Cevians AD , BE , CF of triangle ABC are concurrent iff*

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1.$$

Theorem 2.6 (Menelaus' Theorem). *Points D , E , F on lines BC , CA , AB are collinear iff*

$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = -1 \quad (\text{signed ratios}).$$

2.4 Chapter 4 — Assorted Configurations

Notable configurations to internalize:

- Simson line
- Nine-point circle
- Orthocenter reflection over midpoint lies on circumcircle
- Miquel's theorem
- Spiral similarities

2.5 Chapter 5 — Classical Geometry Revisited

Theorem 2.7 (Ptolemy's Theorem). *For a cyclic quadrilateral $ABCD$,*

$$AC \cdot BD = AB \cdot CD + AD \cdot BC.$$

2.6 Chapter 6 — Inversive Geometry

Definition 2.8. Inversion with center O and radius r maps a point $P \neq O$ to the point P' on ray OP with $OP \cdot OP' = r^2$.

Key facts:

- Lines/circles through O map to lines not through O , and vice versa.
- Circles not through O map to circles not through O .

- Inversion preserves angles (it is conformal).
- Use inversion to kill tangency conditions.

3 Exercises & Problems

Problem 3.1. Let ABC be a triangle with circumcircle ω . The angle bisector from A meets ω again at M . Show $MB = MC$.

Proof. Since AM bisects $\angle BAC$, we have $\angle BAM = \angle CAM$. By the inscribed angle theorem, $\angle BAM = \angle BCM$ and $\angle CAM = \angle CBM$, so $\angle BCM = \angle CBM$, giving $MB = MC$. $\square$

4 Takeaways

- Use directed angles by default to eliminate case splits in circle problems.
- Power of a point + radical axis is the engine of most circle configuration problems.
- Inversion with center at a point of tangency almost always simplifies.
- Ceva and Menelaus are the right tools whenever concurrence/collinearity involves ratios on the sides of a triangle.